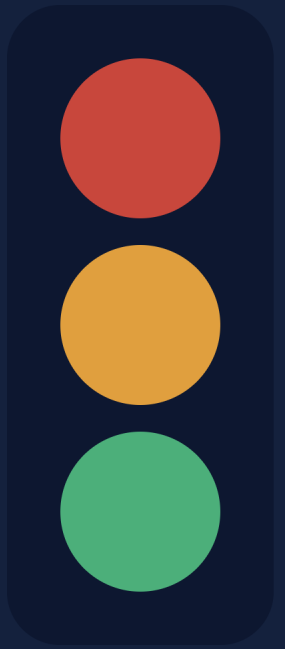
PROJECT ABSTRACT

Traffic Flow Pattern Analysis

Using K-Means and DBSCAN clustering



INTRODUCTION

Problem and aim

The problem

Road sensors record speed, vehicle count and occupancy every few minutes, but nobody labels each reading as free-flow or congested. Studying thousands of readings by hand is slow, and unusual readings are easy to miss.

Our aim

1. Group traffic readings into states: free-flow, moderate, congested
2. Detect unusual readings that may be incidents or sensor faults
3. Do it without labels, using unsupervised learning

DATASET

Traffic flow dataset

1,010
rows (readings)

14
attributes (columns)

5
attributes used

960
rows clustered

Clustering features (3)

speed_kmh, traffic_volume, occupancy

Supporting (2)

timestamp, road_segment (6 roads)

Cleaning

−10 duplicates, −6 impossible values, 34 incomplete rows skipped

METHODOLOGY

From raw data to traffic states

01

Data cleaning

Remove duplicates and impossible values; handle missing values

02

EDA

11 graphs: distributions, correlation, location and time

03

Scaling

StandardScaler gives every feature mean 0 and std 1

04

K-Means

Splits data into K = 3 traffic states; K checked by elbow and silhouette

05

DBSCAN

Finds dense groups; isolated points become noise (outliers)

06

Evaluation

Silhouette, Davies-Bouldin, Calinski-Harabasz; PCA plots